

2. Caesium has many isotopes.
One of these is caesium-137.

(a) The symbol for the isotope caesium-137 is $^{137}_{55}\text{Cs}$.

Tick (✓) the box next to the symbol of another isotope of caesium.

[1]

$^{137}_{54}\text{Cs}$ ☐

$^{133}_{55}\text{Cs}$ ☐

$^{138}_{57}\text{Cs}$ ☐

(b) When caesium-137 decays, beta particles are emitted from unstable nuclei.

(i) Tick (✓) the box next to the correct statement about unstable nuclei.

[1]

They have too many electrons

☐

They have too many protons

☐

The number of neutrons and protons is unbalanced

☐

(ii) Tick (✓) the box next to the correct symbol for a beta particle.

[1]

$^1_{-1}\beta$ ☐

$^0_{-1}\beta$ ☐

$^{-1}_0\beta$ ☐

(iii) Tick (✓) the box next to the correct statement about a beta particle.

[1]

It is a helium nucleus

☐

It is an electromagnetic wave

☐

It is a high energy electron

☐


- (c) The activity of a sample of caesium-137 is 160 units.
It has a half-life of 30 years.

- (i) Underline **one** number in the brackets to complete the sentence. [1]

After 30 years, the activity of the caesium-137 sample will be (**20** / **40** / **80**) units.

- (ii) Jason says the activity of the sample will be zero after 60 years.
Explain whether you agree with Jason. [2]

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